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$$\int (x^2 + 1)(x + 1)^5 dx$$

Stel $x + 1 = t \Rightarrow dx = dt$

$$\begin{aligned} &= \int ((t - 1)^2 + 1)t^5 dt = \int (t^2 - 2t + 2)t^5 dt = \int (t^7 - 2t^6 + 2t^5) dt \\ &= \frac{t^8}{8} - \frac{2t^7}{7} + \frac{2t^6}{6} + C = \frac{(x + 1)^8}{8} - \frac{2(x + 1)^7}{7} + \frac{2(x + 1)^6}{6} + C \\ &= \frac{(x + 1)^8}{8} - \frac{2(x + 1)^7}{7} + \frac{(x + 1)^6}{3} + C \end{aligned}$$

References